

MAD-CORE dataset summary

MAD-CORE is an *ordered coreset* of 2^{24} (≈ 16.8 M) structures picked by farthest-point sampling (FPS) from MAD-COSMO, a pool of 1.7 billion structures collected from 656 public atomistic datasets. Because the rows are stored in FPS order, the first 2^k rows are themselves the level- k coreset. Every energy and force in the record is **predicted by the model used for the selection** (`ecumetric_energy` , `ecumetric_nc_forces`): there are no DFT labels.

This notebook:

1. checks the files against `index.json` ;
2. streams **all** 2^{24} frames once (in parallel, with a small extxyz parser checked against ASE) to summarise sources, composition, sizes, labels and the coreset diagnostics (`epsilon_sq` , `voronoi_population` , ZBL filters);
3. builds **metatomic** `System` s with `vesin.metatomic` neighbour lists for geometry checks, and stores the labels as **metatensor** `TensorMap` s with the layouts `metatomic` defines for `energy` and `non_conservative_force` ;
4. checks the FPS ordering against the shipped selection features;
5. compares the labels with PET-MAD, run directly as a metatomic model;
6. writes a **chemiscope** view of the level-11 coreset (the first 2048 structures), with described properties, force arrows and map settings.

Download (keep the `.gz` files, ASE and this notebook read them directly):

```
MC_TOKEN=<preview-token> bash ~/metawork/etc/download-madcore-extxyz.sh --keep-gz
plus index.json , dataset_metadata.csv and madcore_features_0-2pow18.h5 from the same record.
```

```
data: /home/boittier/data/madcore device: cuda
```

1. Files

`index.json` lists every file with its size, SHA-256 and row range. Set `MADCORE_VERIFY_SHA=1` to hash the local copies (about a minute for the 11 GB of structures).

	file	family	rows	first_row	GB	local	size_ok
	madcore_features_0-2pow18.h5	features	262,144	0	0.455	True	True
	madcore_features_2pow18-2pow19.h5	features	262,144	262,144	0.455	False	False
	madcore_features_2pow19-2pow20.h5	features	524,288	524,288	0.909	False	False
	madcore_features_2pow20-2pow21.h5	features	1,048,576	1,048,576	1.816	False	False
	madcore_features_2pow21-2pow22.h5	features	2,097,152	2,097,152	3.625	False	False
	madcore_features_2pow22-2pow23.h5	features	4,194,304	4,194,304	7.239	False	False
	madcore_features_2pow23-2pow24.h5	features	8,388,608	8,388,608	14.460	False	False
	LICENSES.md	global			0.000	True	True
	README.md	global			0.000	True	True
	dataset_metadata.csv	global			0.000	True	True
	example.py	global			0.000	True	True
	image_flags.npz	global			0.000	True	True
	madcore_0-2pow18.extxyz.gz	xyz	262,144	0	0.098	True	True
	madcore_2pow18-2pow19.extxyz.gz	xyz	262,144	262,144	0.105	True	True
	madcore_2pow19-2pow20.extxyz.gz	xyz	524,288	524,288	0.235	True	True
	madcore_2pow20-2pow21.extxyz.gz	xyz	1,048,576	1,048,576	0.537	True	True
	madcore_2pow21-2pow22.extxyz.gz	xyz	2,097,152	2,097,152	1.231	True	True
	madcore_2pow22-2pow23.extxyz.gz	xyz	4,194,304	4,194,304	2.766	True	True
	madcore_2pow23-2pow24.extxyz.gz	xyz	8,388,608	8,388,608	6.031	True	True

7 structure shards present, 16,777,216 rows (100% of 2^{24})

2. Source datasets

`dataset_metadata.csv` describes the 656 source datasets (grouped into `dataset_group s`), their licence, DOI, and how many structures they contributed to the 1.7 B-structure MAD-COSMO pool.

656 datasets in 455 groups, 1,693,986,898 structures in MAD-COSMO

		datasets
source kind	colabfit	398
	source_shard_and_entry	227
	source_token_or_hash	30
licence	mad_coreset	1
	CC-BY-4.0	532
	MIT	55
	CC0-1	23
	BSD-3-CLAUSE	14
	APACHE-2	14
	NIST-PD	11
	CC-BY-SA-4	3
	CC-BY-3	1
	CC	1
	GDB TERMS OF USE (GDB.UNIBE.CH): CITATION REQUIRED; NO PATENT USE; NO REDISTRIBUTION WITHOUT WRITTEN PERMISSION OF J.-L. REYMOND	1
LICENSE ADDENDUM,CC-BY-4.0		1

3. One pass over every structure

ASE parses about 3k frames/s, which would take hours for 2^{24} frames. The reader below parses the same extxyz at about 30k frames/s per process and runs one process per shard. For every frame it keeps the header metadata and a few derived scalars (so the statistics below cover the *whole* coreset), the composition as a sparse row, and full `ase.Atoms` for the rows we sample. It is checked against `ase.io.read` right after.

```
reader matches ASE on madcore_0-2pow18.extxyz.gz (291 periodic frames checked)
reader matches ASE on madcore_2pow23-2pow24.extxyz.gz (299 periodic frames checked)
```

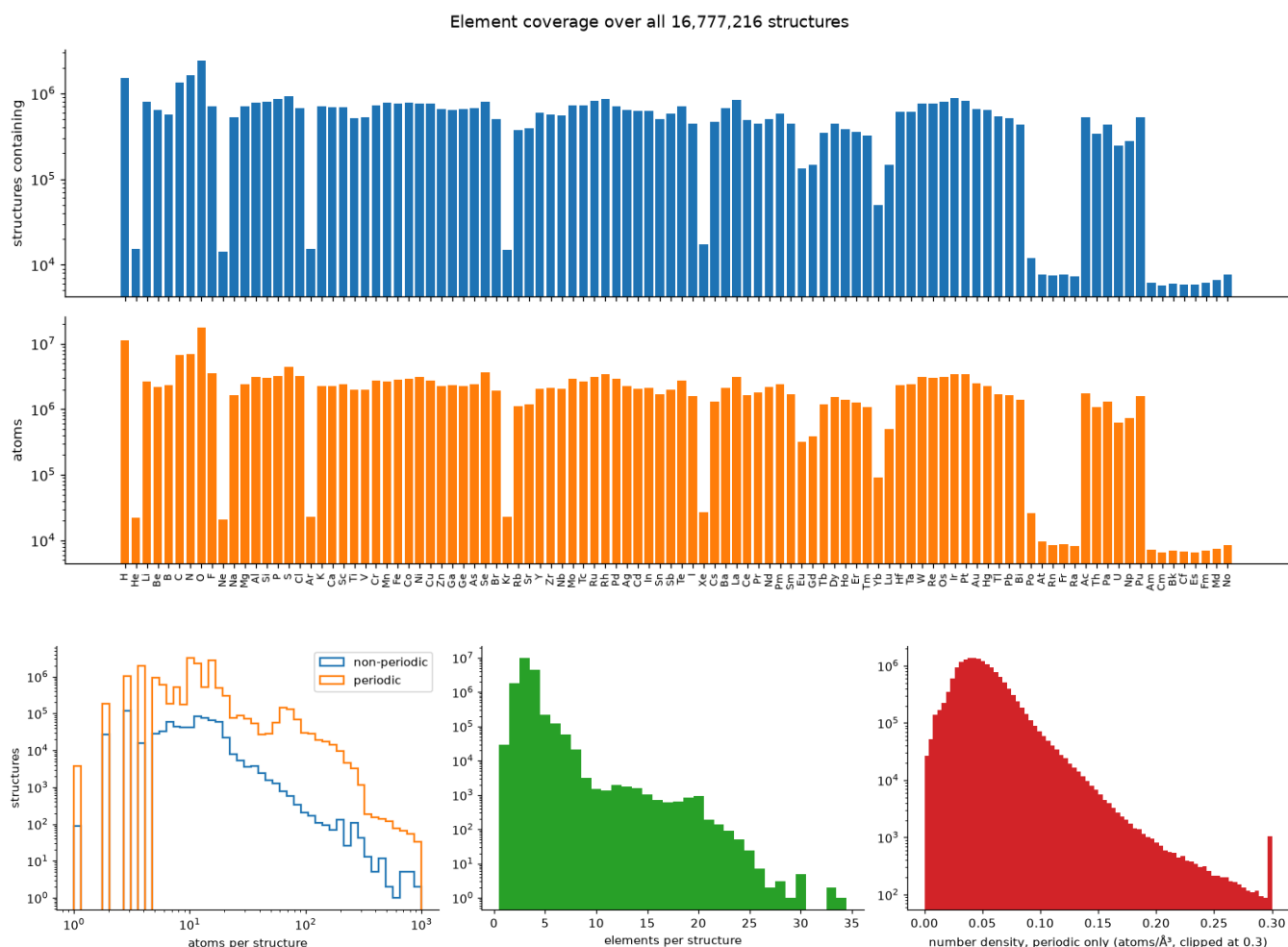
```
16,777,216 structures, 216,185,078 atoms; per-atom layouts: ['species:S:1:pos:R:3:ecu
metric_nc_forces:R:3']
```

4. Composition and size

102 elements: H He Li Be B C N O F Ne Na Mg Al Si P S Cl Ar K Ca Sc Ti V Cr Mn Fe Co
Ni Cu Zn Ga Ge As Se Br Kr Rb Sr Y Zr Nb Mo Tc Ru Rh Pd Ag Cd In Sn Sb Te I Xe Cs Ba
La Ce Pr Nd Pm Sm Eu Gd Tb Dy Ho Er Tm Yb Lu Hf Ta W Re Os Ir Pt Au Hg Tl Pb Bi Po At
Rn Fr Ra Ac Th Pa U Np Pu Am Cm Bk Cf Es Fm Md No
666 dataset_ids in the structures, 3 without an entry in dataset_metadata.csv ['MAD-
1.5', 'mad_next', 'mad_sol']

	structures	atoms (med.)	periodic	first pick	coreset	MAD- COSMO	enrich.
dataset_group							
OMat24	9,452,931	12	100%	1	56.34%	6.25%	9.0
Alexandria	3,159,704	6	100%	3	18.83%	6.98%	2.7
JARVIS_QE_TB	590,179	3	100%	2	3.52%	0.05%	71.8
MP-ALOE	571,477	6	100%	1841	3.41%	0.05%	63.4
JARVIS_OQMD_no_CFID	382,134	4	100%	458	2.28%	0.05%	47.5
nomad	379,008	5	99%	31	2.26%	0.32%	7.1
OMol25	367,234	10	0%	2029	2.19%	8.39%	0.3
mad-next	340,555	3	57%	0	2.03%	0.02%	101.0
JARVIS_CFID_OQMD	299,175	4	100%	170	1.78%	0.03%	65.7
Materials_Project	174,790	14	100%	6	1.04%	0.37%	2.8
MatPES-2025.1	171,580	4	100%	5496	1.02%	0.05%	21.1
Carolina_Materials	164,670	10	100%	1379	0.98%	0.01%	77.6
ANI-1	143,329	16	1%	7621	0.85%	1.47%	0.6
OC25	116,474	122	100%	42427	0.69%	0.45%	1.5
Alex_MP-20	85,959	8	100%	25401	0.51%	0.04%	11.7
Finding_new_crystal_compound...	80,489	8	100%	11860	0.48%	0.01%	37.1
JARVIS_ALIGNN_FF	25,424	4	100%	171	0.15%	0.02%	8.4
UNEP_v1_2023	25,245	32	100%	343	0.15%	0.01%	23.3
OC20_S2EF_train_all	24,466	49	100%	15340	0.15%	7.91%	0.0
QM-22	22,116	7	0%	4923	0.13%	0.06%	2.3
Vector-QM24	18,134	7	0%	27609	0.11%	0.07%	1.7
COLL	15,906	7	0%	31780	0.09%	0.01%	11.5
Matbench	15,267	5	100%	5587	0.09%	0.02%	6.0
HEA25_high_entropy_transitio...	14,164	36	100%	40319	0.08%	0.00%	55.8
HME21	12,538	32	100%	124803	0.07%	0.00%	50.7
ABC2D6-16_PRL_2018	11,129	10	100%	15372	0.07%	0.00%	51.4
OC22-IS2RE	10,167	72	100%	103	0.06%	0.52%	0.1
ANI-2x	8,444	10	0%	28543	0.05%	2.76%	0.0

dataset_group	structures	atoms (med.)	periodic	first pick	coreset	MAD- COSMO	enrich.
mcloudarchive__2daf9-7pk21	7,590	45	100%	83424	0.05%	0.00%	34.8
OC20-IS2RES	5,928	63	100%	5899	0.04%	1.11%	0.0



5. The coreset ordering

`k_set` is the level at which a row entered: level k holds rows $2^{k-1} \dots 2^k - 1$, so the level- k coreset is the first 2^k rows. `epsilon_sq` is the (squared, in scaled feature space) Hausdorff distance of the coreset to the whole pool, i.e. its resolution. Up to level 19 it is a single value per level (the levels were picked as blocks); from level 20 on it decreases pick by pick, so the value at the end of level k (its minimum) is the resolution of the first 2^k rows. Section 9 checks the per-level values against the features. `voronoi_population` is the number of MAD-COSMO structures closest to each pick, i.e. the weight a pick carries. The last panel shows how the mix of sources changes between levels.

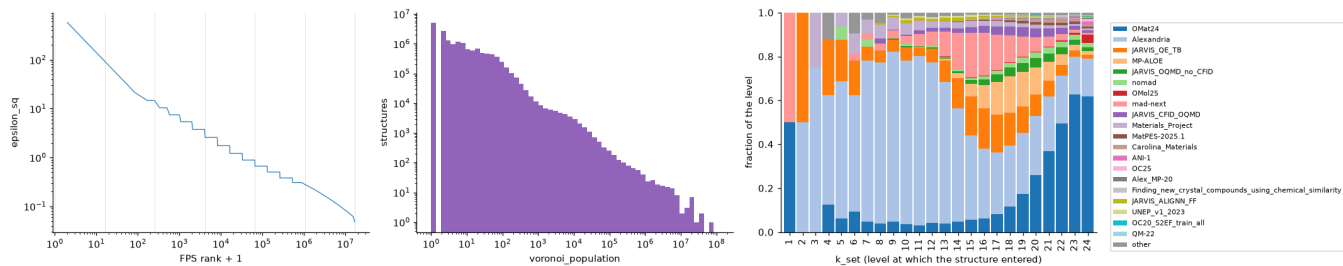
epsilon_sq monotone non-increasing: True

k_set == ceil(log2(fps_order + 1)) except row 0: True | row 0 has k_set 1

levels with a single epsilon_sq: [1, 2, 3, 4, 5, 6, 7, 8, 9, 10, 11, 12, 13, 14, 15, 16, 17, 18, 19]

voronoi populations sum to 1,694,502,355 (MAD-COSMO has 1,693,986,898)

	rows	epsilon_sq	distinct_epsilon	voronoi_population
k_set				
1	2	578.3	1	77,269
2	2	335.4	1	2,625
3	4	157.1	1	871
4	8	85.54	1	1,163
5	16	49.18	1	6,072
6	32	30.4	1	6,156
7	64	21.3	1	1,455
8	128	14.72	1	6,060
9	256	10.42	1	4,760
10	512	7.472	1	4,931
11	1,024	5.377	1	5,501
12	2,048	3.758	1	9,661
13	4,096	2.569	1	15,174
14	8,192	1.74	1	25,610
15	16,384	1.216	1	53,241
16	32,768	0.886	1	157,883
17	65,536	0.6638	1	362,382
18	131,072	0.5035	1	1,616,690
19	262,144	0.3804	1	2,551,963
20	524,288	0.2823	112,219	6,275,294
21	1,048,576	0.2032	444,132	15,817,789
22	2,097,152	0.1406	499,597	37,788,932
23	4,194,304	0.09275	509,472	97,490,362
24	8,388,608	0.04748	680,820	1,532,220,511



6. Labels

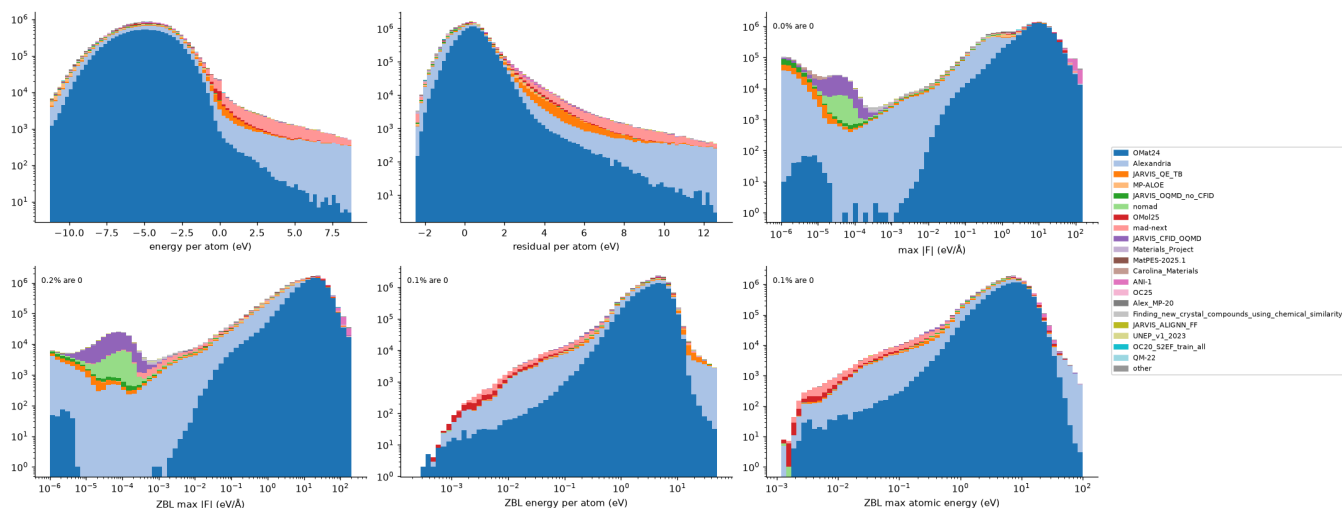
Energies from different sources are on different references (and every label here comes from the same model anyway), so we fit a per-element linear baseline $E \approx \sum_Z n_Z \mu_Z$ over all structures (the composition model metatrain fits before training) and look at the residual per atom. The ZBL diagnostics were used to filter unphysical close contacts when the coreset was built.

per-element energies μ (eV)

H:	He:	Li:	Be:	B:	C:	N:	O:	F:	Ne:	Na:	Mg:
-3.96	0.75	-2.15	-3.01	-5.95	-8.44	-8.36	-7.91	-5.06	0.12	-1.26	-1.26
Al:	Si:	P:	S:	Cl:	Ar:	K:	Ca:	Sc:	Ti:	V:	Cr:
-3.85	-5.71	-5.81	-5.16	-3.06	0.34	-0.12	-2.29	-5.38	-7.23	-8.43	-8.39
Mn:	Fe:	Co:	Ni:	Cu:	Zn:	Ga:	Ge:	As:	Se:	Br:	Kr:
-8.17	-6.62	-6.73	-5.46	-3.31	-1.09	-3.24	-4.65	-4.97	-4.40	-2.09	2.09
Rb:	Sr:	Y:	Zr:	Nb:	Mo:	Tc:	Ru:	Rh:	Pd:	Ag:	Cd:
-0.45	-2.39	-5.57	-7.79	-9.69	-9.84	-9.34	-8.22	-6.29	-3.74	-1.93	-0.21
In:	Sn:	Sb:	Te:	I: -1.22	Xe:	Cs:	Ba:	La:	Ce:	Pr:	Nd:
-2.36	-3.90	-4.02	-3.35		1.95	-0.47	-1.97	-5.47	-6.43	-5.65	-5.20
Pm:	Sm:	Eu:	Gd:	Tb:	Dy:	Ho:	Er:	Tm:	Yb:	Lu:	Hf:
-4.57	-3.97	-4.14	-4.27	-3.07	-1.64	-2.56	-2.35	-2.17	-1.72	-5.21	-7.80
Ta:	W:	Re:	Os:	Ir:	Pt:	Au:	Hg:	Tl:	Pb:	Bi:	Po:
-9.45	-9.57	-9.85	-9.51	-8.55	-6.07	-2.93	0.44	-1.80	-2.96	-3.52	-2.61
At:	Rn:	Fr:	Ra:	Ac:	Th:	Pa:	U:	Np:	Pu:	Am:	Cm:
0.28	6.10	3.82	3.46	-4.31	-7.66	-9.16	-9.82	-9.09	-7.21	-1.43	-0.24
Bk:	Cf:	Es:	Fm:	Md:	No:						
0.72	1.42	2.12	2.71	2.78	3.27						

composition baseline residual: median $|r|$ 0.565, MAE 0.740, RMSE 1.122 eV/atom -- a coreset keeps the far tails, hence mean $\gg$ median

group	medians per group				structures
	E/atom (eV)	residual (eV)	max F (eV/Å)	rms F (eV/Å)	
OMat24	-5.08	0.389	8.61	4.43	9,452,931
Alexandria	-4.84	-0.383	0.409	0.273	3,159,704
JARVIS_QE_TB	-4.13	0.429	4.12e-06	3.06e-06	590,179
MP-ALOE	-5.69	0.327	1.44	1.01	571,477
JARVIS_OQMD_no_CFID	-5.5	-0.347	1.17e-06	7.77e-07	382,134
nomad	-4.99	-0.0463	8.4e-05	5.53e-05	379,008
OMol25	-5.18	0.666	6.15	3.35	367,234
mad-next	-2.69	0.829	1.91	1.41	340,555
JARVIS_CFID_OQMD	-4.81	0.0451	1.36e-05	9.18e-06	299,175
Materials_Project	-6.13	-0.502	1.42	0.881	174,790
MatPES-2025.1	-6.36	-0.344	1.28	0.913	171,580
Carolina_Materials	-4.88	0.375	0.115	0.09	164,670
ANI-1	-4.2	1.89	86.4	33.8	143,329
OC25	-5.41	-0.189	7.34	1.95	116,474
Alex_MP-20	-4.69	-0.839	0.206	0.136	85,959
Finding_new_crystal_compound...	-5.93	-0.791	0.279	0.191	80,489
JARVIS_ALIGNN_FF	-4.11	0.0582	0.352	0.246	25,424
UNEP_v1_2023	-6.64	-0.359	1.74	0.878	25,245
OC20_S2EF_train_all	-5.03	-0.287	4.16	1.28	24,466
QM-22	-4.56	1.25	15.4	8.89	22,116
other	-5.58	0.232	2.9	1.42	200,277



Outliers

Structures whose residual is more than 8 robust standard deviations from their group's median, or whose largest force exceeds 100 eV/Å. In a coreset these are expected to be over-represented (FPS favours the edges of the distribution), so treat the list as "worth a look", not "wrong".

113,006 flagged structures (0.674%)

flagged	
dataset_group	
ANI-1	53299
OMat24	25463
Alexandria	23385
OC20-IS2RES	2004
Materials_Project	1244
OC22-IS2RE	1199
JARVIS_CFID_OQMD	1024
QM-22	898
UNEP_v1_2023	719
OMol25	606
OC22-S2EF	485
JARVIS_QE_TB	427
Si_Al_Ti_Seko_PRB_2019	410
OC20_S2EF_train_all	232
JARVIS_ALIGNN_FF	184

	fps_order	dataset_group	n_atoms	energy_per_atom	residual_per_atom	max_force
2770328	2770328	OMat24	30	-5.088625	2.007295	156.911960
2907189	2907189	OC22-IS2RE	87	-6.629766	0.326835	156.447582
11108067	11108067	OMat24	60	-4.072766	2.388131	155.912156
9121155	9121155	OC22-S2EF	113	-5.385176	1.787451	155.772102
2051178	2051178	OC22-S2EF	90	-6.501505	0.546363	155.529303
7776390	7776390	Materials_Project	49	-1.851767	4.836095	155.404880
3890217	3890217	OMat24	46	-3.564151	2.112773	155.372564
5017272	5017272	OMat24	64	-5.037971	1.862267	155.147084
13694454	13694454	OMat24	84	-3.938613	1.909574	155.071160
3505901	3505901	OMat24	98	-3.547657	2.825442	155.064981

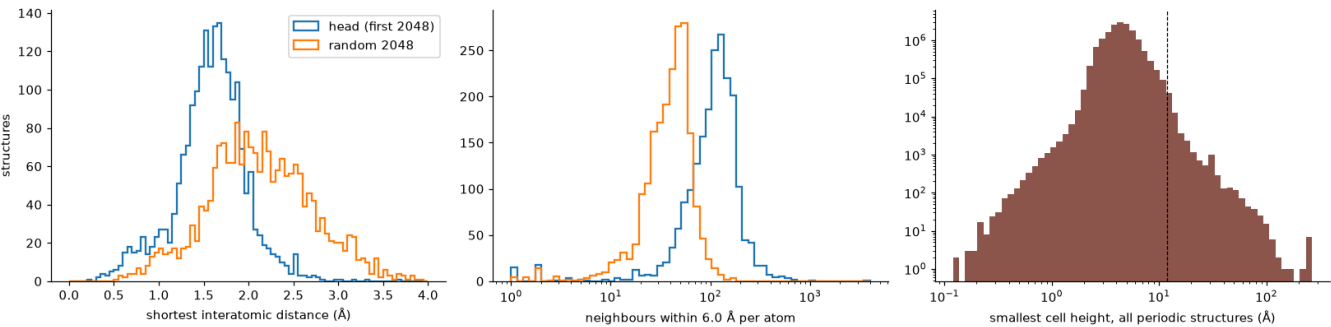
7. metatomic systems and neighbour lists

The head of the coreset and the random sample are converted to metatomic `System`s (float64, Å) and given a half neighbour list from `vesin.metatomic`, exactly as an engine would before calling a model. The shortest pair distance flags overlapping atoms; the minimum cell height over the whole dataset tells us whether small periodic cells need a neighbour list with multiple images (vesin handles that).

```
System with 1 atoms, non periodic
TensorBlock
  samples (0): ['first_atom', 'second_atom', 'cell_shift_a', 'cell_shift_b', 'cell_shift_c']
  components (3): ['xyz']
  properties (1): ['distance']
  gradients: None
```

99.7% of periodic cells are thinner than 2 x cutoff

	fps_order	dataset_group	n_atoms	min_distance	max_force
609	609	Alexandria	12	0.236431	58.791814
1109	1109	Alexandria	14	0.334289	105.230728
61	61	Alexandria	14	0.334423	92.762416
611	611	Alexandria	10	0.351055	70.224712
1123	1123	Alexandria	10	0.372303	62.757756
122	122	Alexandria	9	0.380852	92.197625
348	348	OMat24	5	0.396356	66.883072
8	8	OMat24	9	0.400026	42.959023



8. Labels as metatensor `TensorMap`s

The labels are stored with the metadata metatomic prescribes for the `energy` and `non_conservative_force` outputs. The forces here are *direct* (non-conservative) predictions, so they belong in their own per-atom `TensorMap`, **not** in the `positions` gradient of the energy. Keeping the sample `system` equal to `fps_order` makes any slice of the coreset self-describing.

```
TensorBlock
  samples (2048): ['system']
  components (): []
  properties (1): ['energy']
  gradients: None
```

```
TensorBlock
  samples (16826): ['system', 'atom']
  components (3): ['xyz']
  properties (1): ['non_conservative_force']
  gradients: None
```

|sum of forces| over the head: median 6.76e-02, max 1.03e+02 eV/Å

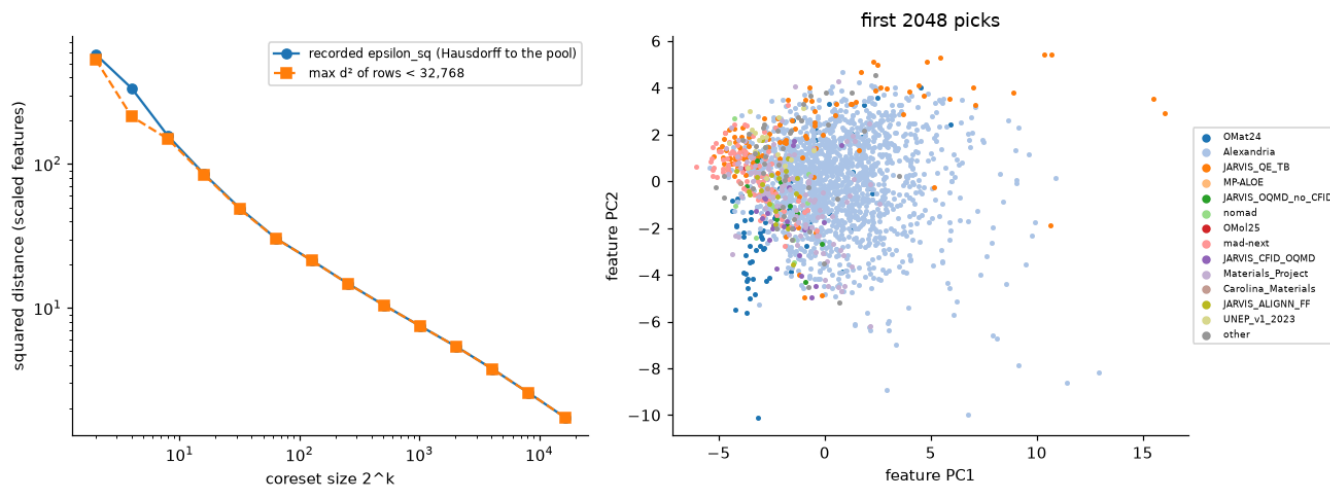
9. The coreset in feature space

`madcore_features_0-2pow18.h5` holds the 512-dimensional features used for the selection. The head's scaled features are stored as a metatensor `TensorMap` and projected with PCA for the map. We then check the recorded resolution: since the first 2^{18} rows are a subset of the pool, the largest squared distance from any of them to the level- k coreset must not exceed `epsilon_sq[k]`, and for a good coreset it should come close. The last line tests whether the order is also *greedy* point by point (each pick being the farthest point from all earlier ones).

```
{'features': (262144, 512), 'mean': (512,), 'scale': ()}
```

	epsilon_sq	max d ² of later rows	d ² of next pick	covered	greedy
k					
1	578.3	532.7	532.7	True	True
2	335.4	213.6	213.6	True	True
3	157.1	149.6	149.6	True	True
4	85.54	84.38	84.38	True	True
5	49.18	48.44	31.2	True	False
6	30.4	30.34	25.26	True	False
7	21.3	21.28	17.6	True	False
8	14.72	14.69	13.59	True	False
9	10.42	10.42	9.579	True	False
10	7.472	7.47	7.206	True	False
11	5.377	5.375	5.355	True	False
12	3.758	3.758	3.752	True	False
13	2.569	2.569	2.569	True	True
14	1.74	1.74	1.739	True	True

every later row inside epsilon_sq of each level: True; greedy at 6/14 levels
 PCA explained variance: [0.312 0.174 0.095]



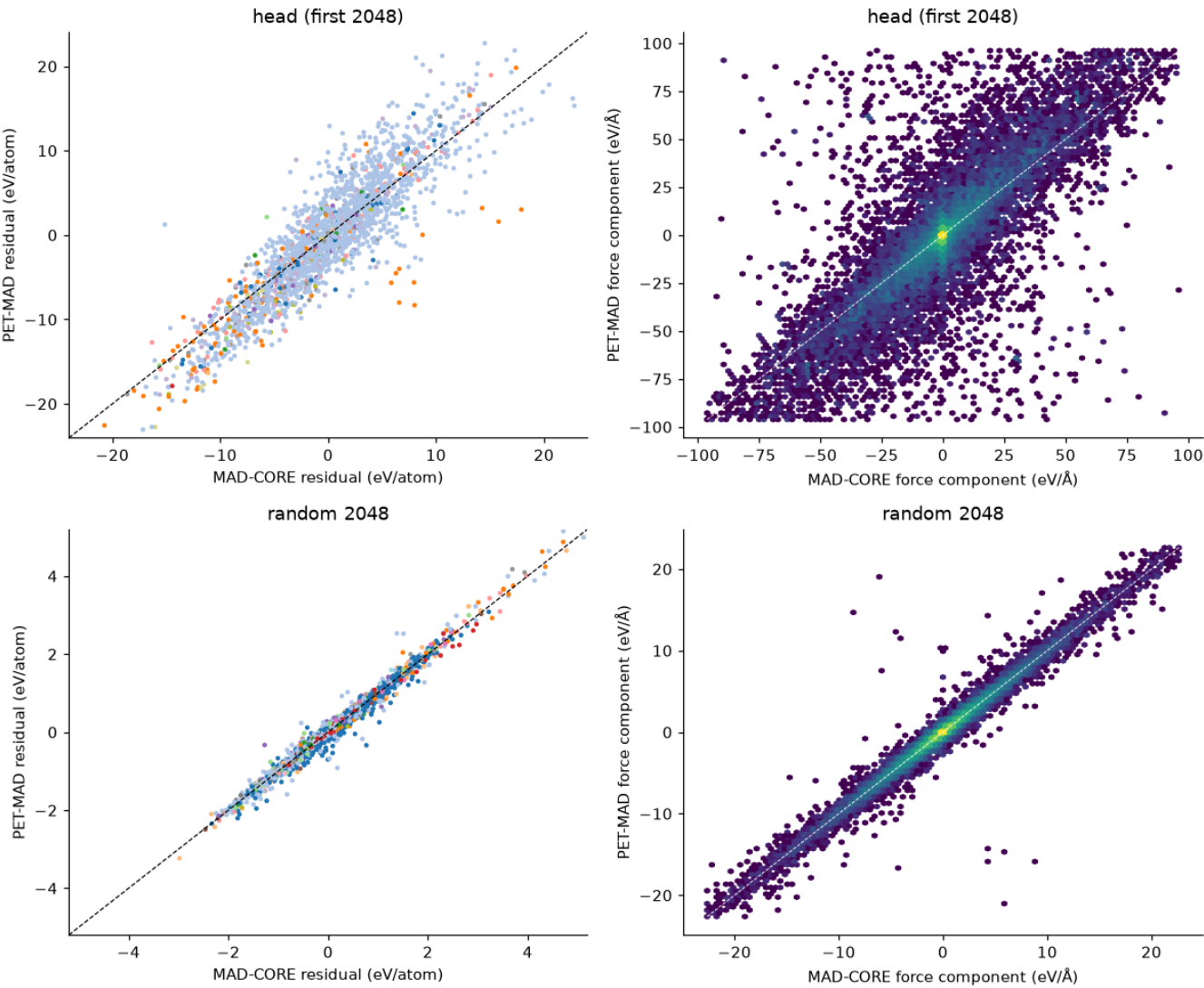
10. Comparison with PET-MAD

PET-MAD (here the XS model) is evaluated directly through the metatomic `AtomisticModel` interface, requesting its `energy` and `non_conservative_force` outputs with the neighbour lists the model asks for (checked against `MetatomicCalculator` on a few structures). Both the FPS head (the most extreme structures) and the random sample (typical ones) are compared. Since

reference energies differ, each model's own composition baseline is removed first. Note that MAD-CORE forces are exactly zero on high-symmetry structures, where PET-MAD's direct (not exactly equivariant) forces are only approximately zero.

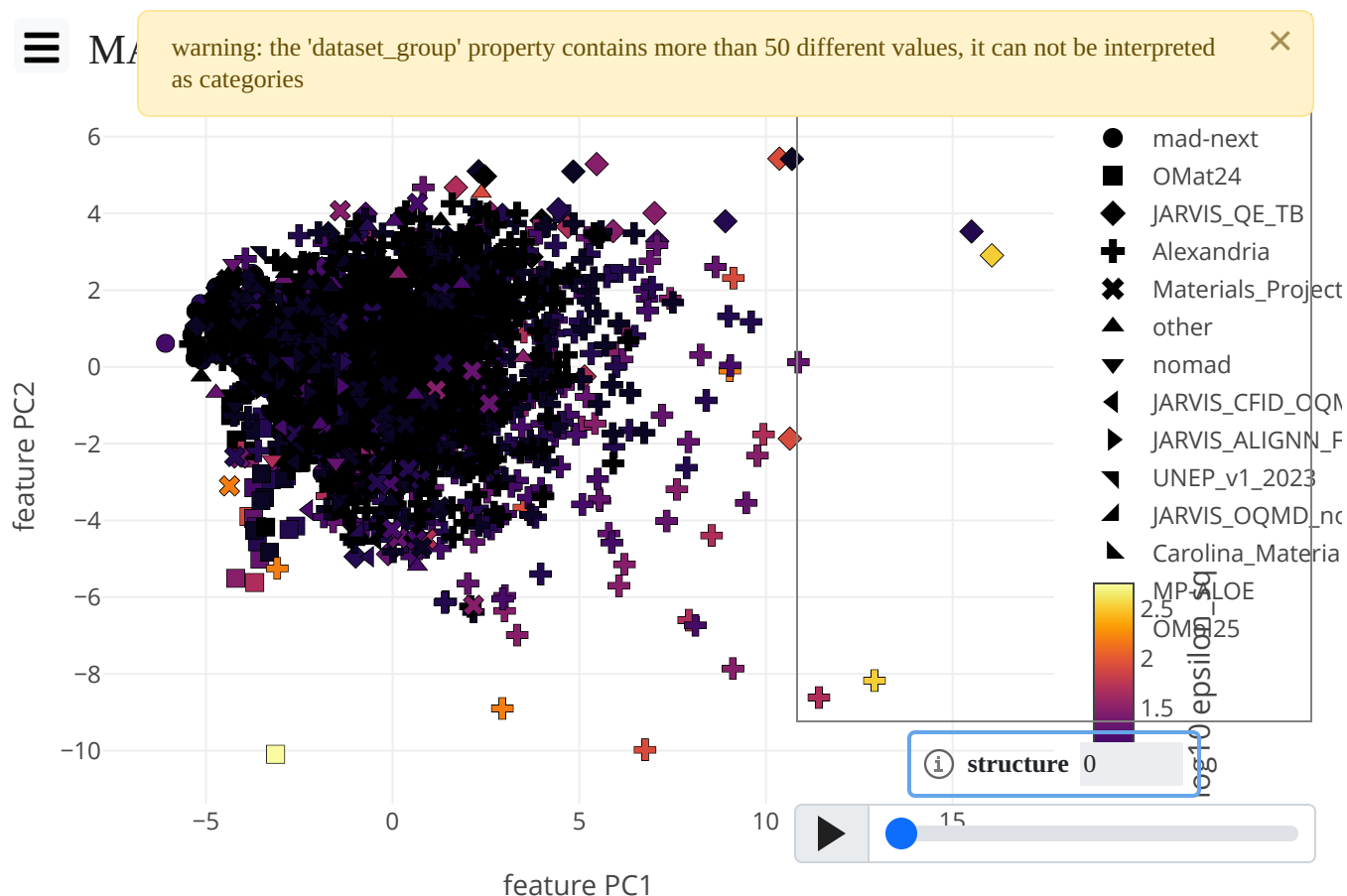
```
[W923 10:55:11.872270658 model.cpp:1114] Warning: Current metatomic version (0.2.0-dev1125+git.fe7ef0a) is not compatible with the version (0.1.17) used to export the model at '/home/boittier/metawork/openmm-metatomic/models/pet-mad-xs-v1.5.0.pt'; proceed at your own risk. (function check_atomistic_model)
```

	head (first 2048)	random 2048
structures	2.05e+03	2.05e+03
energy residual MAE (eV/atom)	2.6	0.0975
energy residual median AE (eV/atom)	1.96	0.0627
force MAE (eV/Å)	8.19	0.293
force median AE (eV/Å)	1.95	0.166
atoms with MAD-CORE $ F < 1e-4$	0.0813	0.048



11. Interactive chemiscope view of the level-11 coreset

Properties are passed explicitly with descriptions and units rather than dumped from `atoms.info`, forces are drawn as arrows, and the input is written to `madcore-outputs/madcore-head.json.gz`, which opens on <https://chemiscope.org> or with `chemiscope.show_input` without re-running anything. (The record also ships `madcore_0-2pow20.chemiscope.json.gz`, a view of the first 2^{20} structures.)



MAD-CORE	
structures	16,777,216
atoms	216,185,078
elements	102
source datasets / groups present	666 / 336
periodic	95.6%
atoms per structure (median / mean / max)	10 / 12.9 / 1000
largest group	OMat24 (56.3%)
labels (model predictions, no DFT)	ecumetric_energy, ecumetric_nc_forces
composition-baseline residual, median r / MAE (eV/atom)	0.565 / 0.740
median max F (eV/Å)	3.95
flagged outliers	113,006 (0.674%)
coreset resolution epsilon_sq, level 11 / last level	5.38 / 0.0475
shortest distance in samples (Å)	0.24